

HAZARDOUS AREA CLASSIFICATION

Equipment, Instruments and Electrical Systems

Basic to Advanced Practical Guide for Oil & Gas EPC Projects

FOUNDATION

Zones, groups, temperature

APPLICATION

Equipment and instrument review

ADVANCED

Release, ventilation, packages

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Purpose and use of this manual

This manual provides a practical basic-to-advanced learning path for hazardous area classification, with emphasis on how classification affects mechanical equipment, instruments, electrical equipment and packaged skids in oil and gas EPC projects.

ENGINEERING LIMITATION This is training and preliminary decision support, not an approved hazardous-area classification study. Project classification must be completed or endorsed by competent Process Safety, Electrical, Instrumentation, Mechanical and Package Engineering personnel using project-approved standards and operating data.

Learning outcomes

- Explain why and how hazardous areas are classified.
- Differentiate IEC Zones from the North American Division system.
- Read Ex equipment markings and certificates.
- Select and review electrical, instrument and mechanical equipment against an area requirement.
- Identify package-level interfaces, common nonconformities and information needed before a final decision.

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1. What is hazardous area classification means

A hazardous area is a three-dimensional space where flammable gas, vapour, mist or combustible dust may form an explosive atmosphere in sufficient quantity that special precautions are required to prevent ignition.

Four questions answered by HAC

1. Where can an explosive atmosphere occur?
2. How frequently and for how long can it occur?
3. What substance may be present?
4. What equipment protection level is required?

CORE IDEA Hazardous area classification establishes the location requirement. Equipment certification and installation demonstrate how an item avoids becoming an effective ignition source.

The explosion triangle

Element	Meaning	Examples
Fuel	Gas, vapour, mist or dust	Hydrogen, methane, hydrocarbon vapour, combustible powder
Oxidant	Normally oxygen in air	Normal atmospheric air
Ignition source	Energy capable of ignition	Spark, arc, hot surface, friction, static, flame, reaction

Hazardous-area design normally assumes that fuel and air may coexist. Equipment and installation are therefore selected so that they do not become an effective ignition source.

2. Classification is for the location

Area classification

The plant location is classified by considering:

- Material handled
- Release source, frequency and release rate
- Physical and ignition properties
- Ventilation and dispersion
- Geometry, confinement and congestion
- Operating and foreseeable abnormal conditions

Equipment selection

After the area is classified, equipment is selected for:

- Zone or Division
- Equipment Protection Level, EPL
- Gas or dust group
- Temperature class or maximum surface temperature
- Ambient-temperature range
- Protection concept and certification scheme
- Environmental and installation conditions

VENDOR REVIEW RULE The statement “suitable for hazardous area” is incomplete. Require the complete Ex marking, certificate, schedule and all special conditions of use.

3. IEC Zone system

Gas and vapour Zones

Zone	Definition	Typical interpretation
Zone 0	Present continuously, for long periods or frequently	Inside vessels, tanks or continuously exposed analyser chambers
Zone 1	Likely to occur occasionally in normal operation	Around normal vents, sampling points, seals or routine releases
Zone 2	Unlikely in normal operation and, if present, persists briefly	Around normally tight items where release is abnormal and limited

The Zone reflects the probability and duration of an explosive atmosphere, not directly the consequence severity of an explosion.

Dust Zones

Zone	Meaning
Zone 20	Explosive dust cloud present continuously, for long periods or frequently
Zone 21	Explosive dust cloud likely occasionally in normal operation
Zone 22	Explosive dust cloud unlikely in normal operation or present briefly

Dust layers can insulate equipment and raise its surface temperature, become resuspended as an explosive cloud, and extend the hazardous area if housekeeping is inadequate.

4. North American Class/Division system

System	Higher-hazard location	Lower-hazard location
Class I gas/vapour	Division 1	Division 2
IEC/API Zone system	Zone 0 or Zone 1	Zone 2

CAUTION Division 1 is not automatically equivalent to Zone 1. It may encompass circumstances corresponding to Zone 0 or Zone 1. Apply the governing project rules rather than relabelling the area.

API RP 500 addresses Class I, Division 1 and Division 2 petroleum facilities. API RP 505 addresses Class I, Zone 0, Zone 1 and Zone 2. Do not mix systems within a package unless the project deliberately permits dual certification and defines the installation rules.

5. How a hazardous area is classified

Step 1: Identify flammable substances

- Gas or vapour composition
- Flash point, boiling point and vapour pressure

- Lower and upper flammable limits
- Relative vapour density
- Auto-ignition temperature
- Operating pressure and temperature
- Potential to flash, aerosolize or form mist
- Dust cloud and layer properties, where applicable

IMPORTANT DISTINCTION A liquid is not classified merely because it is called hydrocarbon. The concern is whether operating or release conditions can generate an ignitable vapour, gas or mist. High-flash-point liquids released under pressure may create a flammable mist.

Step 2: Identify sources of release

- Pump and compressor seals
- Valve stems and flanged joints
- Instrument process connections
- Drain, vent and sample points
- Relief-valve outlets and tank vents
- Chemical injection pump packing
- Filter or strainer opening points
- Loading connections and analyser vents
- Seal-oil reservoirs and skid drain funnels

Not every flange produces the same hazardous extent. The adopted project method determines whether and how a connection is treated under normal or abnormal conditions.

Step 3: Determine grade of release

Grade	Meaning	Examples
Continuous	Continuous, frequent or long duration	Tank vapour space; continuously emitting vent
Primary	Periodic or occasional in normal operation	Routine sample, open drain, normal vent
Secondary	Not expected normally; infrequent and brief	Normally tight flange, stem or instrument connection

Release grade influences, but does not alone determine, the Zone. Ventilation and dilution must also be assessed.

Step 4: Estimate release rate

- Opening size and discharge coefficient
- Upstream pressure and fluid temperature
- Gas molecular mass or liquid density
- Vapour pressure and flash fraction
- Choked or non-choked flow
- Release direction and impingement
- Obstructions, congestion and two-phase behaviour

Advanced studies calculate releases from credible physical scenarios rather than selecting an arbitrary standard hole size.

Step 5: Assess ventilation and dilution

Ventilation can be natural, mechanically supplied, locally extracted or provided by a purge/pressurization system. Assess both availability and effectiveness.

- Open air does not guarantee good dispersion.

- Acoustic enclosures, walls, decks, pits and congestion can reduce dilution.
- Heavier-than-air vapours may collect at low points.
- Lighter gases may accumulate near roofs if escape paths are restricted.

Step 6: Determine Zone and extent

The final output includes both a Zone designation and a three-dimensional physical extent. The envelope may be spherical, cylindrical, conical, a full enclosure, a region above or below a source, or a shape modified by barriers.

NO COPY-PASTE DISTANCES Extent distances must come from the project-approved IEC, API, NFPA, client or dispersion-modelling method. Do not reuse a distance from another project without confirming the substance, pressure, release opening and ventilation basis.

6. Equipment groups

Gas equipment: Group II

Gas group	Relative ignition severity	Representative reference gas
IIA	Least stringent of the three	Propane
IIB	More stringent	Ethylene
IIC	Most stringent	Hydrogen or acetylene

Equipment certified for IIC can generally meet IIB and IIA group requirements, subject to the complete certificate and installation restrictions. The reverse is not acceptable. This hierarchy concerns ignition transmission and ignition energy, not process consequence severity.

Dust equipment: Group III

Dust group	Material
IIIA	Combustible flyings
IIIB	Non-conductive dust
IIC	Conductive dust

7. Temperature class

Electrical or mechanical equipment must not develop a surface temperature capable of igniting the atmosphere.

Temperature class	Maximum equipment surface temperature
T1	450°C
T2	300°C
T3	200°C
T4	135°C
T5	100°C
T6	85°C

SELECTION RULE If an area requires T4, equipment marked T4, T5 or T6 may be suitable; T3 is not. A higher T-number means a lower and more restrictive surface temperature.

Dust temperature

Dust equipment usually states an actual maximum surface temperature, for example T135°C. Evaluate dust-cloud and layer ignition temperatures, layer thickness, orientation, housekeeping, external heating and insulating effects.

8. Equipment Protection Levels

Atmosphere	Zone	Normally required EPL	Conceptual protection
Gas	Zone 0	Ga	Very high
Gas	Zone 1	Gb	High
Gas	Zone 2	Gc	Enhanced
Dust	Zone 20	Da	Very high
Dust	Zone 21	Db	High
Dust	Zone 22	Dc	Enhanced

Correct EPL alone is insufficient. The equipment may still be unsuitable because of the wrong group, temperature class, ambient range, gland, certificate conditions, materials or installation.

9. Common electrical protection concepts

Marking	Protection concept	Common applications
Ex d	Flameproof enclosure	Motors, switches, solenoids, local panels
Ex e	Increased safety	Motors, terminal boxes, junction boxes
Ex i	Intrinsic safety	Transmitters, switches, sensors, low-energy circuits
Ex p	Pressurization	Control panels, analyser houses, large enclosures
Ex m	Encapsulation	Solenoid coils, sensors, electronic modules
Ex n	Legacy Zone 2 type of protection	Some legacy Zone 2 equipment
Ex t	Dust protection by enclosure	Motors, instruments and boxes in dust areas
Ex o	Oil immersion	Specialized equipment
Ex q	Powder filling	Specialized electrical devices

What the concepts do

- Ex d permits an internal explosion but prevents transmission outside the enclosure.
- Ex e prevents arcs, sparks and excessive temperatures under defined conditions.
- Ex i limits electrical energy so ignition cannot occur under specified faults.
- Ex p maintains a protective pressure or purge condition so an explosive atmosphere does not enter or persist.

The concepts are different engineering solutions. One is not universally better than another.

10. Instrumentation selection

Instrument review

5. Verify Zone and EPL.
6. Verify gas group or dust group.
7. Verify temperature class or maximum surface temperature.
8. Check ambient-temperature range.
9. Confirm protection concept.
10. Review certificate number, revision and full schedule.
11. Resolve every X special condition and any U component limitation.
12. Verify IP rating, process connection, materials, gland, plug and adaptor.

Intrinsically safe loops

An Ex i transmitter is not automatically a safe loop. Review the complete system: field instrument, cable, terminals, barrier or isolator, input card and earthing arrangement.

Field equipment	Associated apparatus	System check
Ui, li, Pi	Uo, Io, Po	Associated-apparatus output must not exceed field-device input limits
Ci, Li	Co, Lo	Device plus cable capacitance and inductance must remain within permitted limits

- Verify segregation and identification.
- Verify shielding and earthing.
- Use the correct cable parameters and installed length.
- Check the complete loop drawing and barrier channel identification.

Common instrumentation mistakes

- Reviewing only the transmitter rather than the complete IS loop
- Ignoring certificate X conditions
- Using uncertified adaptors or stopping plugs
- Assuming IP66 means explosion protected
- Mixing IS and non-IS wiring improperly
- Ignoring analyser vent discharge location
- Using a weatherproof junction box as if it were Ex certified

11. Electrical-equipment selection

Motors

- Zone/EPL, gas group and temperature class
- Starting and stalled-rotor heating
- Variable-frequency-drive suitability
- Ambient range, duty and service factor
- Space-heater and terminal-box protection
- Glands, stopping plugs, earthing and bonding
- Bearing sensors, winding RTDs and certificate conditions
- Heat transfer from the driven equipment

PACKAGE BOUNDARY A certified motor does not certify the complete pump or compressor package.

Junction boxes and local panels

- Certified enclosure and terminals
- Maximum current and internal power dissipation
- Internal component certification
- Glands, stopping plugs and thread compatibility
- Creepage, clearance and earthing
- Pressurization/purge logic where applicable

Lighting, sockets and heaters

- Maximum surface temperature
- Lamp or heating-element type and wattage
- Portable equipment classification
- Plug/socket interlocking
- Certification of internal heat-producing components
- Heat-tracing design, controls and alarm philosophy

12. Mechanical and rotating equipment

Hazardous-area requirements do not stop at the electrical driver. Pumps, compressors, fans, agitators, couplings, gearboxes and brakes can become ignition sources.

Potential ignition source	Examples
Hot surfaces	Bearings, casing, brakes, lubrication failure
Mechanical sparks	Rubbing, foreign-object impact, coupling contact
Static electricity	Belts, non-conductive materials, fluid movement
Compression/reaction	Adiabatic compression, exothermic reactions
Process release interaction	Seal leakage onto a hot surface

Rotating-equipment review questions

- Can dry running create an unsafe temperature?
- Is loss of lubrication detected and tripped?
- Is bearing temperature monitored?
- Can rotating and stationary parts contact?
- Are belts conductive and bonded?
- Could seal failure release material onto a hot surface?
- Can process casing temperature exceed the area limit?
- Is a non-electrical ignition-hazard assessment required?
- Has the complete package been assessed, or only the driver?

Static mechanical equipment

A static vessel or pipe normally does not need Ex certification merely because it is in a hazardous area. Still assess hot surfaces, insulation, static charge, linings, attached rotating components, lightning/bonding, process reactions, relief releases and electrical instruments.

13. Reading an Ex nameplate

Gas example: Ex db eb IIC T4 Gb

Marking element	Meaning
Ex	Explosion-protected equipment
db	Flameproof protection, level b
eb	Increased-safety protection, level b
IIC	Most demanding Group II gas subgroup
T4	Maximum surface temperature associated with T4
Gb	High EPL, normally suitable for Zone 1

Dust example: Ex tb IIIC T135°C Db

Marking element	Meaning
Ex tb	Dust ignition protection by enclosure, level b
IIIC	Conductive-dust group
T135°C	Marked maximum surface temperature
Db	High EPL, normally suitable for Zone 21

Certificate suffixes

- X: special conditions of use apply. These are mandatory engineering conditions.
- U: the item is an Ex component and is not necessarily complete equipment ready for independent installation.

14. Packaged skid example

Assumed installation requirement: Zone 2, IIB, T3, maximum ambient 50°C.

Item	Required review
Pump motor	At least Gc, IIB or IIC, T3 or safer, and ambient suitable for 50°C
Local junction box	Correct EPL, group, temperature, terminals and glands
Pressure transmitter	Correct marking; if Ex i, complete loop verification
Solenoid valve	Coil certification, group, T-class and ambient rating
Vibration sensor	Sensor and associated barrier/monitor compatibility
Space heater	Included in motor certificate; controlled against excessive temperature
Mechanical seal	Release source included in HAC; leakage routed safely
Pump and coupling	Mechanical ignition-hazard assessment where required
Skid drain	Confirm open/closed design and additional release source
Local panel	Ex-certified assembly or approved pressurized/purged solution
Cable glands	Match concept, cable type, thread and certificate

Item	Required review
Hot piping	Surface temperature compatible with area requirement
Documentation	Certificates, drawings and special conditions reconciled

KEY LESSON The area requirement flows through process design, equipment design, instrument architecture, electrical selection, layout, ventilation, controls, installation, inspection and maintenance.

15. Internal versus external hazardous areas

- A vessel may be Zone 0 internally.
- External areas around openings may be Zone 1 or Zone 2.
- External surfaces elsewhere may be non-hazardous.
- An analyser cabinet may contain hazardous process tubing, have a classified interior under leakage, or use an approved purge system.
- The analyser vent outlet may create a separate external hazardous region.

Show equipment location, vent routing and enclosure boundaries clearly on the hazardous-area drawing.

16. Ventilation failure

For a mechanically ventilated room or enclosure, define the classification with ventilation operating and failed, together with the required safeguards.

- Ventilation reliability and monitoring
- Differential-pressure or flow alarm
- Trip or power isolation
- Start-up purge and time delay
- Gas detection and voting philosophy
- Emergency operation philosophy

DESIGN RULE Do not treat ventilation as permanently available unless availability, monitoring and failure response are demonstrated.

17. Gas detection does not normally erase the Zone

Gas detection is normally a protective layer rather than a substitute for correct classification of a release source. It can provide warning, activate ventilation, initiate shutdown or isolation and support emergency response.

Any use of gas detection to support a classification change requires the governing standard, project philosophy, coverage, voting logic, availability, response time and approving authority to permit it.

18. Required project documents

Document group	Typical documents
Process inputs	PFD, P&ID, heat and material balance, composition, equipment and line lists, operating/design conditions, relief/vent study, drains and sampling philosophy
HAC deliverables	Classification philosophy, area schedule, drawings, source-of-release register, calculations, ventilation assessment, property table, assumptions and deviations

Document group	Typical documents
Equipment documents	Datasheets, instrument index, motor/load lists, Ex schedule, certificates and schedules, GA drawings, gland schedules, IS calculations, purge calculations, ignition-hazard assessment

19. Vendor-document review checklist

No.	Review field	Status / comment
1	Tag number	
2	Installation location	
3	Area classification	
4	Zone or Division	
5	Gas/dust group	
6	Temperature class / maximum surface temperature	
7	Required EPL	
8	Actual equipment marking	
9	Ambient range	
10	Certificate scheme and number	
11	Certificate revision	
12	X conditions	
13	U component status	
14	Protection concept	
15	Cable entry type and thread	
16	Gland certification	
17	IP rating	
18	Process rating	
19	Compliance status	
20	Deviation or clarification required	

Red flags

- “ATEX suitable” without full marking or certificate schedule
- Certificate does not cover the precise model
- Ambient range below the project maximum
- Motor certified but accessories are not
- IIB equipment offered for an IIC requirement
- T3 equipment offered where T4 is required
- Gc offered for Zone 1
- U component presented as complete equipment
- X conditions not addressed
- Mismatched Ex d cable-entry devices

- Non-Ex items located inside the classified extent
- Division markings offered on a Zone-only project
- Datasheet, GA, MR and certificate information conflict

20. Engineering document hierarchy

Establish the project-specific contractual precedence before deciding whether equipment complies. Unless the project defines another hierarchy, review requirements in these logical groups:

Group	Documents
A. Contractual and project-specific	Purchase Order, MR, project specifications, approved datasheets, scope, design basis, client requirements, approved clarifications and deviations
B. Company requirements	Company standards, specifications, engineering practices and technical instructions
C. External requirements	Laws, regulations, adopted codes, standards and statutory authority requirements
D. Vendor information	Offer, datasheets, drawings, calculations, procedures, manuals and sub-vendor documents

CONFLICT CONTROL If two requirements conflict, record both documents, revisions and clauses; explain the technical and contractual conflict; apply precedence only where documented; otherwise obtain contractual, client or lead-engineer clarification.

Example compliance decision

Requirement: Zone 1, IIC, T4, Gb. Vendor offer: Ex db IIB T3 Gb.

Parameter	Required	Offered	Finding
EPL	Gb	Gb	Acceptable for Zone 1
Gas group	IIC	IIB	Not acceptable
Temperature	T4	T3	Not acceptable

The appropriate action is replacement or a formally assessed deviation, not an assumption that Ex d alone is sufficient.

21. Information Required Before Final Decision

- Governing country and statutory requirements
- Project-approved HAC standard and edition
- Area-classification drawing and revision
- Zone/Division, group and temperature requirement
- Minimum and maximum ambient temperature
- Fluid composition and ignition properties
- Source-of-release and ventilation basis
- Full equipment marking
- Certificate and complete schedule
- Special conditions of use
- Process design conditions
- Package layout and release locations
- Electrical installation and gland philosophy

- Mechanical ignition-hazard assessment
- Contractual document precedence

EDITION CONTROL A newly published standard is not automatically the contractual project edition. Any edition change requires project document-control and technical-governance review.

22. Fast review model: Z-G-T-E-A-C-I

Letter	Check	Question
Z	Zone / Division	Is the item suitable for the classified location?
G	Gas / dust group	Does the certification meet or exceed the required group?
T	Temperature	Is the class or marked surface temperature acceptable?
E	EPL	Does the protection level match the Zone?
A	Ambient	Does the certificate cover project ambient conditions?
C	Certificate	Is the precise model covered, including X/U conditions?
I	Installation	Are glands, cable, earthing, segregation and assembly correct?

MEMORY AID If any one of Z-G-T-E-A-C-I is wrong, the equipment may be unsuitable even when an Ex symbol is present.

23. Recommended learning progression

Level	Topics
Foundation	Explosion triangle; Zones and Divisions; release grades; groups; temperature classes; EPLs
Intermediate	Protection concepts; nameplates; IS loops; motors and boxes; mechanical ignition sources; glands
Advanced	Release calculations; ventilation/dilution; extent calculation; mist/two-phase releases; pressurization; IS system calculations; package assemblies; brownfield changes; failure cases; inspection, repair and maintenance

MAIN TAKEAWAY Hazardous area classification identifies where and how an explosive atmosphere may occur. Equipment certification demonstrates how a particular item avoids becoming an ignition source. Safe design requires both to match, together with correct installation, inspection and maintenance.

Appendix A. Worked review examples

A1. Instrument transmitter

Area requirement: Zone 1, IIC, T4, ambient -10°C to +55°C. Offered marking: Ex ia IIC T4 Ga, certified ambient -40°C to +60°C.

Check	Result	Comment
Zone/EPL	Pass	Ga is suitable for the required Zone 1 application
Gas group	Pass	IIC matches
Temperature	Pass	T4 matches
Ambient	Pass	Certificate range covers project range
Certificate	Open	Review model coverage, schedule and X conditions
Installation	Open	Complete IS loop and cable/entity calculation required

A2. Pump package

Area requirement: Zone 2, IIB, T3. The motor is marked Ex ec IIC T3 Gc, but the coupling guard is non-conductive and the skid includes a non-certified local switch.

Element	Preliminary finding
Motor	Marking appears compatible, subject to certificate and ambient review
Coupling/guard	Mechanical and electrostatic ignition risks require assessment
Local switch	Non-compliant unless outside the classified extent or otherwise suitably protected
Package	Not acceptable on motor certification alone

Appendix B. Reference framework

Use the project-approved editions. The following official publications formed the technical reference framework for this learning manual:

Reference	Scope / use
IEC 60079-10-1:2020	Explosive atmospheres - Part 10-1: Classification of areas - Explosive gas atmospheres. IEC Webstore.
IEC 60079-10-2:2026	Explosive atmospheres - Part 10-2: Classification of areas - Explosive dust atmospheres. IEC Webstore.
IEC 60079-0:2026	Explosive atmospheres - Part 0: Equipment - General requirements. IEC Webstore.
IEC 60079-14:2024	Explosive atmospheres - Part 14: Electrical installation design, selection and installation of equipment, including initial inspection. IEC Webstore.
IECEx Standards	IECEx list of standards used for Ex equipment, installations, inspection, repair, intrinsic safety and gas detection.
ISO 80079-36:2016	Explosive atmospheres - Part 36: Non-electrical equipment for explosive atmospheres - Basic method and requirements. ISO.
API RP 500, 4th edition	Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Division 1 and Division 2. API.
API RP 505	Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1 and Zone 2. API.

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EQUIPMENT PROTECTION LEVELS

Ga, Gb, Gc, Da, Db, Dc, Ma and Mb

Table-Based Engineering Guide for Equipment, Instruments, Electrical Systems and Packaged Skids

Ga ZONE 0 VERY HIGH	Gb ZONE 1 HIGH	Gc ZONE 2 ENHANCED
Da ZONE 20 VERY HIGH	Db ZONE 21 HIGH	Dc ZONE 22 ENHANCED

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Purpose and limitations

This guide provides a concise, table-based explanation of all IEC Equipment Protection Levels and their normal relationship to gas, dust and mining hazardous areas. It is intended for engineering training, technical bid evaluation, vendor-document review and preliminary package compliance screening.

ENGINEERING LIMITATION EPL is only one selection parameter. Final acceptance must also verify the classified area, protection concept, gas or dust group, temperature class or maximum surface temperature, ambient range, complete certificate, special conditions and installation requirements. Approval remains with the responsible engineering and project authorities.

How to read an EPL

Position	Letter	Meaning
First character	G	Explosive gas, vapour or mist atmosphere
First character	D	Explosive combustible-dust atmosphere
First character	M	Mine susceptible to firedamp
Second character	a	Very high equipment protection
Second character	b	High equipment protection
Second character	c	Enhanced equipment protection

MEMORY RULE G = Gas, D = Dust, M = Mine. The suffix a provides the highest protection, b the next level and c the normal minimum level for the least hazardous surface Zone.

1. Master EPL selection table

Atmosphere	Area	Minimum EPL	Protection level	Ignition-performance principle	Normally suitable for
Gas, vapour or mist	Zone 0	Ga	Very high	Not an ignition source in normal operation, expected malfunctions or rare malfunctions. Exact fault criteria depend on the protection concept.	Zones 0, 1 and 2
Gas, vapour or mist	Zone 1	Gb	High	Not an ignition source in normal operation or expected malfunctions.	Zones 1 and 2
Gas, vapour or mist	Zone 2	Gc	Enhanced	Not an ignition source in normal operation. Additional abnormal-condition provisions depend on the concept.	Zone 2
Combustible dust	Zone 20	Da	Very high	Very high protection against ignition of dust cloud or layer during normal operation and defined malfunction conditions.	Zones 20, 21 and 22
Combustible dust	Zone 21	Db	High	High protection against ignition during normal operation and expected malfunctions.	Zones 21 and 22
Combustible dust	Zone 22	Dc	Enhanced	Protection against ignition principally during normal operation.	Zone 22
Mine susceptible to firedamp	Mining area	Ma	Very high	Very high mining protection; continued energization may be permitted where the governing mining requirements allow.	Mining applications requiring Ma
Mine susceptible to firedamp	Mining area	Mb	High	High mining protection; equipment is normally de-energized when an explosive atmosphere is detected.	Mining applications requiring Mb

Normal surface-industry hierarchy: Ga > Gb > Gc for gas and Da > Db > Dc for dust. The applicable protection-concept standard controls the detailed construction, testing and malfunction requirements.

2. Gas EPL selection

Zone-to-EPL relationship

Gas area	Minimum normal EPL	Higher EPL also normally acceptable	Normally unacceptable
Zone 0	Ga	None above Ga	Gb, Gc
Zone 1	Gb	Ga	Gc
Zone 2	Gc	Gb, Ga	Non-Ex equipment unless the design specifically permits it

Gas acceptance matrix

Equipment EPL	Zone 0	Zone 1	Zone 2
Ga	ACCEPT	ACCEPT	ACCEPT
Gb	NO	ACCEPT	ACCEPT
Gc	NO	NO	ACCEPT

Detailed comparison

Criterion	Ga	Gb	Gc
Protection level	Very high	High	Enhanced
Normal corresponding Zone	Zone 0	Zone 1	Zone 2
Normal operation	Protected	Protected	Protected
Expected malfunction	Protected	Protected	Concept-dependent; not the general defining basis
Rare malfunction	Protected	Not generally required	Not generally required
Typical intrinsic safety	Ex ia	Ex ib	Ex ic
Typical encapsulation	Ex ma	Ex mb	Ex mc
Lower-hazard Zone use	Zones 1 and 2	Zone 2	No lower gas Zone

3. EPL Ga

Item	Explanation
Atmosphere	Gas, vapour or mist
Protection level	Very high
Normal minimum Zone	Zone 0
Normal Zone suitability	Zones 0, 1 and 2
Example marking	Ex ia IIC T4 Ga

Ga equipment is normally suitable for Zones 0, 1 and 2. It provides the highest surface-industry gas EPL and is intended to remain non-incendive during normal operation, expected malfunctions and rare malfunctions. The exact design and fault criteria depend on the certified protection concept.

Typical applications

- Tank probes exposed to vessel vapour space
- Low-energy level sensors
- Certain transmitters and optical equipment
- Special combined-protection assemblies

PRACTICAL NOTE A tank-mounted instrument can cross Zone boundaries. The probe may require Ga because it enters a Zone 0 vapour space even when the external transmitter body is in Zone 1 or Zone 2.

4. EPL Gb

Item	Explanation
Atmosphere	Gas, vapour or mist
Protection level	High
Normal minimum Zone	Zone 1
Normal Zone suitability	Zones 1 and 2
Example marking	Ex db IIB T4 Gb

Gb equipment is normally suitable for Zones 1 and 2, but not Zone 0. It is intended to remain non-incendive during normal operation and expected malfunctions.

Typical applications

- Zone 1 motors
- Junction boxes and local stations
- Transmitters, solenoids and detectors
- Lighting and package auxiliaries

PACKAGE NOTE A Gb motor does not certify the complete pump or compressor package. Review all instruments, switches, heaters, glands and mechanical ignition sources.

5. EPL Gc

Item	Explanation
Atmosphere	Gas, vapour or mist
Protection level	Enhanced
Normal minimum Zone	Zone 2
Normal Zone suitability	Zone 2 only
Example marking	Ex ec IIC T3 Gc

Gc equipment is normally suitable for Zone 2 only. It is principally intended to remain non-incendive in normal operation; concept-specific requirements may address additional defined conditions.

Typical applications

- Zone 2 motors
- Local indicators and electronic modules
- Lighting and junction boxes
- Package control components

COMMON ERROR IIC gas group and T4 temperature class do not make Gc equipment acceptable in Zone 1. EPL must independently meet the Zone requirement.

6. Dust EPL selection

Zone-to-EPL relationship

Dust area	Minimum normal EPL	Higher EPL also normally acceptable	Normally unacceptable
Zone 20	Da	None above Da	Db, Dc
Zone 21	Db	Da	Dc
Zone 22	Dc	Db, Da	Non-Ex equipment unless the design specifically permits it

Dust acceptance matrix

Equipment EPL	Zone 20	Zone 21	Zone 22
Da	ACCEPT	ACCEPT	ACCEPT
Db	NO	ACCEPT	ACCEPT
Dc	NO	NO	ACCEPT

Detailed comparison

Criterion	Da	Db	Dc
Protection level	Very high	High	Enhanced
Normal corresponding Zone	Zone 20	Zone 21	Zone 22
Typical enclosure concept	Ex ta	Ex tb	Ex tc
Typical occurrence	Continuous/frequent dust cloud	Occasional in normal operation	Unlikely and brief
Lower-hazard Zone use	Zones 21 and 22	Zone 22	No lower dust Zone
Temperature marking	Actual maximum surface temperature	Actual maximum surface temperature	Actual maximum surface temperature

DUST WARNING EPL alone is not sufficient. Verify IIIA/IIIB/IIIC group, marked maximum surface temperature, dust-cloud and layer ignition properties, layer thickness, ingress protection, ambient range and housekeeping assumptions.

7. Typical protection concepts by EPL

The following table is for learning and preliminary screening. Never assign an EPL solely from the protection-concept letters. Use the complete equipment marking and certificate.

EPL	Typical concepts	Example marking	Common examples
Ga	Ex ia; Ex ma; suitable combined or optical concepts	Ex ia IIC T4 Ga	Tank probes, sensors and low-energy devices
Gb	Ex db; Ex eb; Ex ib; Ex mb; suitable pressurization, oil immersion or powder filling	Ex db IIB T4 Gb	Motors, boxes, instruments, solenoids and local stations

EPL	Typical concepts	Example marking	Common examples
Gc	Ex ec; Ex ic; Ex mc; Ex pzc; applicable Ex dc; certain legacy Ex n concepts	Ex ec IIC T3 Gc	Zone 2 motors, instruments, lighting and electronics
Da	Ex ia; Ex ma; Ex ta	Ex ta IIIC T120°C Da	Equipment in continuously dusty process spaces
Db	Ex ib; Ex mb; Ex tb; suitable pressurization	Ex tb IIIC T135°C Db	Zone 21 motors, boxes, instruments and lighting
Dc	Ex ic; Ex mc; Ex tc	Ex tc IIIB T100°C Dc	Zone 22 equipment
Ma	Mining concepts certified to Ma	Certificate-specific	Essential mining monitoring or communications
Mb	Mining concepts certified to Mb	Certificate-specific	Mining equipment subject to de-energization

KEY DISTINCTION EPL states how much ignition protection is provided. The protection concept states how that protection is achieved.

8. Practical equipment-selection examples

Area requirement	Offered marking	EPL result	Other critical check	Preliminary finding
Zone 0, IIC, T4	Ex ia IIC T4 Ga	Meets	Certificate, IS entity data, ambient and installation	Potentially acceptable
Zone 0, IIC, T4	Ex ib IIC T4 Gb	Fails	Other parameters cannot compensate for EPL	Not acceptable
Zone 1, IIB, T4	Ex db IIB T4 Gb	Meets	Ambient, certificate, gland and installation	Potentially acceptable
Zone 1, IIB, T4	Ex ec IIC T4 Gc	Fails	IIC and T4 do not compensate for Gc	Not acceptable
Zone 2, IIB, T3	Ex db IIC T4 Gb	Exceeds	Check certificate, ambient and installation	Potentially acceptable
Zone 2, IIC, T4	Ex ec IIB T4 Gc	Meets	IIB does not meet IIC	Not acceptable
Zone 21, IIIC, T135°C	Ex tb IIIC T120°C Db	Meets	Check dust layer basis and ambient	Potentially acceptable
Zone 21, IIIC, T135°C	Ex tc IIIC T100°C Dc	Fails	Temperature cannot compensate for EPL	Not acceptable
Zone 22, IIIB, T135°C	Ex tb IIIC T120°C Db	Exceeds	Check certificate and installation	Potentially acceptable

9. EPL and ATEX category mapping

Atmosphere	EPL	Normal Zone	ATEX category
Gas	Ga	Zone 0	1G
Gas	Gb	Zone 1	2G
Gas	Gc	Zone 2	3G
Dust	Da	Zone 20	1D
Dust	Db	Zone 21	2D
Dust	Dc	Zone 22	3D
Mining	Ma	Mining	M1
Mining	Mb	Mining	M2

This mapping is useful for screening, but final regulatory acceptance still requires review of the complete marking, conformity-assessment route, certificate and project jurisdiction.

10. Complete EPC review checklist

Check	Required information	Typical rejection condition
Zone	Zone 0, 1, 2, 20, 21 or 22	Equipment EPL below the Zone requirement
EPL	Ga/Gb/Gc or Da/Db/Dc	Gc for Zone 1; Dc for Zone 21
Protection concept	Ex d, e, i, p, m, t, etc.	Concept unsuitable for equipment or installation
Group	IIA/IIB/IIC or IIIA/IIIB/IIIC	IIB offered for an IIC requirement
Temperature	T1-T6 or marked dust temperature	T3 offered where T4 is required
Ambient range	Project minimum and maximum	Certificate limited to +40°C; project requires +55°C
Certificate	Required scheme and full schedule	Precise model not covered
X condition	Special conditions of use	Condition not incorporated in design
U suffix	Ex component status	Component presented as complete equipment
Cable entry	Gland, cable and thread details	Gland concept or thread mismatch
Package assembly	All electrical, instrument and mechanical items	Only the motor is compliant
Installation	Earthing, segregation, barriers and inspection	Installation conflicts with certificate instructions

11. Fast review model: Z-G-T-E-A-C-I

Code	Review item	Decision question
Z	Zone or Division	Is the equipment suitable for the classified location?
G	Gas or dust group	Does certification meet or exceed the required group?
T	Temperature	Is the class or maximum surface temperature acceptable?
E	Equipment Protection Level	Does the EPL match or exceed the Zone requirement?
A	Ambient range	Does the certificate cover project ambient conditions?
C	Certificate	Is the exact model covered, including X and U conditions?
I	Installation	Are glands, cable, barriers, earthing and assembly correct?

FINAL REVIEW RULE Correct EPL answers only one question: whether the equipment provides the required broad level of ignition protection. It never replaces review of the complete Ex marking, certificate and installation.

12. Information Required Before Final Decision

- Approved hazardous-area classification drawing and revision
- Governing jurisdiction, project specification and adopted standard edition
- Zone or Division and required EPL
- Gas or dust subgroup
- Temperature class or permitted maximum surface temperature
- Project minimum and maximum ambient temperature
- Complete equipment marking
- Certificate and full certificate schedule
- X special conditions and U component status
- Cable, gland, barrier, segregation and earthing details
- Package layout and internal/external hazardous-area boundaries
- Purchase Order, Material Requisition and approved deviations
- Multidisciplinary confirmation by Process Safety, Electrical, Instrument, Mechanical and Package Engineering as applicable

13. Document hierarchy and reference framework

Requirement precedence

Group	Documents to review
A. Contractual/project-specific	Purchase Order, MR, project specifications, approved datasheets, design basis, client requirements and approved deviations
B. Company	Company standards, specifications, practices and technical instructions
C. External	Applicable law, regulation, adopted code, standard and authority requirement
D. Vendor	Offer, datasheets, drawings, calculations, procedures, manuals and sub-vendor documents

CONFLICT CONTROL Where requirements conflict, identify both documents, revisions and clauses. Apply precedence only when documented; otherwise obtain contractual, client or lead-engineer clarification.

Reference framework

Publisher / publication	Use in this guide
IEC 60079-0:2026	General requirements, construction, testing, marking and Equipment Protection Level framework
IEC 60079-10-1:2020	Classification of areas containing explosive gas atmospheres
IEC 60079-10-2:2026	Classification of areas containing explosive dust atmospheres
IEC 60079-14:2024	Electrical installation design, selection, erection and initial inspection
IECEx standards list	Protection-concept, installation, inspection, repair and system standards
ISO 80079-36:2016	Non-electrical equipment, ignition-hazard assessment and requirements
API RP 500 / API RP 505	North American Division and Zone approaches for petroleum facilities
CSA C22.1:21, Table 18	User-provided visual reference for equipment suitable for explosive atmospheres; use a licensed copy for Canadian-code application

This is original explanatory material. Standard titles and short designations are included for identification only. Consult properly licensed copies of all governing standards for normative requirements.

CLASS, DIVISION, GROUP AND ZONE

Practical Hazardous-Area Classification Guide

How to understand, compare and apply Division and Zone systems in oil and gas EPC projects

DIVISION SYSTEM

Class + Division + Group + T-class

ZONE SYSTEM

Zone + Gas/Dust Group + T-class + EPL

Prepared for equipment, package and vendor-document review | Revision 0 | 09 September 2026

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NAVIGATION NOTE The sections that follow are maintained in their logical engineering sequence. Headings continue naturally without forcing every section onto a separate page.

Purpose, decision and limitations

This manual explains the North American Class/Division system and the IEC/API/NEC Zone system in a project-application format. It is intended to help package, mechanical, electrical and instrument engineers interpret hazardous-area drawings, review equipment markings, evaluate vendor deviations and identify when two classification descriptions are broadly similar.

ENGINEERING LIMITATION The comparisons are learning and preliminary selection guidance, not automatic code conversions or design approval. Final classification and equipment acceptance must follow the project Design Basis, hazardous-area drawings, Purchase Order, Material Requisition, client requirements, adopted code, statutory requirements and Authority Having Jurisdiction.

The exact decision addressed

Decision element	Question
Hazard type	Does gas/vapour, dust or fibres/flyings create the hazard?
Occurrence	How often and for how long can the explosive atmosphere exist?
Material group	Which gas or dust subgroup applies?
Equipment protection	Which construction, EPL and certification are required?
Cross-use	Can Division-rated equipment be used in a Zone location, or vice versa?
Project acceptance	Do code permission and contractual certification requirements both permit the offered equipment?

1. The essential difference

Division-system expression

Class + Division + Group + Temperature Class
 Example: Class I, Division 1, Group B, T4

Zone-system expression

Zone + Gas/Dust Group + Temperature Class + EPL
 Example: Zone 1, IIC, T4, Gb

API RP 500 addresses petroleum locations classified as Class I, Division 1 and Division 2. API RP 505 addresses petroleum locations classified as Class I, Zone 0, Zone 1 and Zone 2. These are complementary approaches, but the drawings, equipment markings and installation rules must follow the system adopted by the project.

FIRST PROJECT RULE Establish the governing system before reviewing any vendor offer. Do not allow the vendor to select or mix the classification basis after Purchase Order award.

2. Meaning of the terms

Term	Question answered	Division system	Zone system
Class	What general type of combustible material creates the hazard?	Yes	Class I may be retained in NEC/CSA Zone terminology
Division	How likely is the hazardous atmosphere to exist?	Division 1 or 2	No
Zone	How frequently and for how long can the atmosphere exist?	No	Zones 0/1/2 or 20/21/22
Group	Which gas, vapour or dust has relevant ignition properties?	A/B/C/D or E/F/G	IIA/IIB/IIC or IIIA/IIIB/IIIC
Temperature class	What maximum equipment surface temperature is permitted?	Yes	Yes
EPL	What broad ignition-protection level is provided?	Not the primary Division designation	Ga/Gb/Gc or Da/Db/Dc

KEY DISTINCTION Class identifies the material family. Division or Zone identifies occurrence. Group identifies material ignition characteristics. T-class controls hot surfaces. EPL identifies equipment protection level.

3. Class means the hazardous-material type

In the North American system, Class does not indicate severity. Class identifies the broad nature of the combustible material.

Class	Hazardous material	Typical examples
Class I	Flammable gases and flammable- or combustible-liquid-produced vapours capable of forming an ignitable atmosphere	Natural gas, hydrogen, LPG, gasoline vapour, condensate vapour and solvent vapour
Class II	Combustible dust	Sulphur, coal, coke and some chemical powders
Class III	Ignitable fibres or flyings not normally suspended in air in quantities sufficient to form an ignitable mixture	Fibres, flyings and some textile or wood-processing materials

DO NOT CONFUSE Class I does not mean Zone 1, Division 1, IEC Group I or “highest risk.” Class I means a gas or vapour type of hazard.

Complete Division example:
Class I, Division 1, Group D, T3

4. Division means probability of occurrence

Class I, Division 1

Division 1 generally covers locations where ignitable gas or vapour concentrations can exist during normal operating conditions, can occur frequently because of repair, maintenance or leakage, or can be released by equipment breakdown that may simultaneously make electrical equipment an ignition source.

- Inadequately ventilated hydrocarbon pump rooms
- Areas around open hydrocarbon sampling points
- Areas around normal process vents
- Open-transfer connections for flammable liquids
- Enclosures containing instruments that normally vent flammable gas

- Process locations where release occurs in normal operation

Class I, Division 2

Division 2 generally covers locations where flammable gas or vapour is normally confined in closed equipment or piping and may escape only through accidental rupture, leakage or abnormal operation; where adequate positive mechanical ventilation normally prevents accumulation but failure is considered; or where the location is adjacent to Division 1 and vapour communication is possible.

- Outdoor closed hydrocarbon systems
- Normally tight valves, seals, flanges and instrument connections
- Areas surrounding a Division 1 envelope
- Ventilated analyser or process enclosures where ventilation failure is considered

5. Zone means frequency and duration

Gas and vapour Zones

Zone	Occurrence of explosive atmosphere	Simple memory
Zone 0	Present continuously, for long periods or frequently	Continuous/frequent
Zone 1	Likely to occur occasionally during normal operation	Expected occasionally
Zone 2	Unlikely in normal operation and, if present, exists only briefly	Abnormal and brief

Dust Zones

Zone	Occurrence of explosive dust atmosphere
Zone 20	Present continuously, for long periods or frequently
Zone 21	Likely to occur occasionally during normal operation
Zone 22	Unlikely in normal operation and, if present, exists briefly

ZONE OUTPUT An approved hazardous-area classification provides both the Zone designation and the three-dimensional physical extent around each source of release.

6. When the two systems are similar

Division-system condition	Closest Zone-system condition	Why broadly similar	Limitation
Class I, Division 1	Zone 0 or Zone 1	Gas/vapour may be present during normal or frequent conditions	Division 1 does not separate continuous/frequent occurrence from occasional normal-operation occurrence
Class I, Division 2	Zone 2	Both commonly address unlikely, abnormal and short-duration releases	Boundaries, equipment permission and wiring methods remain code-dependent
Class I	Gas/vapour Zones 0, 1 or 2	Both describe gas/vapour hazards	Class I identifies material type, not occurrence frequency
Class II, Divisions 1/2	Dust Zones 20/21/22	Both address combustible dust	No universal one-to-one Division-to-Zone conversion
Class III	Group IIIA fibres/flyings, conceptually	Both can address combustible fibres or flyings	Definitions and installation rules differ

MOST IMPORTANT RULE Division 1 approximately spans Zone 0 plus Zone 1. Therefore, Division 1 is not equal to Zone 1.

Why Division 1 is not equal to Zone 1

Scenario	Zone-system result	Division-system result
Inside a hydrocarbon vessel where vapour is frequent or long-duration	Likely Zone 0	Division 1 concept
Around a normal hydrocarbon vent with occasional release	Likely Zone 1	Division 1 concept

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Division 1
├─ Zone 0 type conditions
└─ Zone 1 type conditions
  
```

When Division 2 and Zone 2 are closest

They are most similar when a closed outdoor process system is not expected to release an ignitable atmosphere during normal operation, but limited abnormal leakage can occur for a short duration and dispersion is adequate. Even then, use the project code to determine classification extent and equipment acceptance.

7. Group means the material subdivision

Class I groups in the Division system

Division group	Representative material	General ignition challenge
Group A	Acetylene	Special and highly demanding
Group B	Hydrogen is a common representative	Very demanding
Group C	Ethylene is a common representative	Intermediate
Group D	Propane is a common representative	Less demanding of the four Class I groups

Group boundaries are based on ignition-related characteristics such as maximum experimental safe gap and minimum igniting-current ratio. Equipment must be certified for the specific required group.

IEC gas subgroups

IEC gas subgroup	Representative examples	Relative equipment requirement
IIA	Propane, petrol vapour, ethanol and acetone examples	Least demanding of IIA/IIB/IIC
IIB	Ethylene/ethene examples	More demanding
IIC	Hydrogen, acetylene and carbon disulphide examples	Most demanding

```

Normal equipment coverage hierarchy:
IIC equipment → IIC, IIB and IIA
IIB equipment → IIB and IIA
IIA equipment → IIA only
  
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Approximate gas-group comparison

Division group	Representative	Broad Zone-group relationship
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Division group	Representative	Broad Zone-group relationship
A	Acetylene	IIC
B	Hydrogen	Generally IIC
C	Ethylene	Generally IIB
D	Propane	Generally IIA

NO AUTOMATIC CONVERSION IIC contains representative gases found in both Division Groups A and B. Use the actual gas, project code and equipment certificate rather than approving by a general comparison table.

IEC equipment Groups

IEC equipment group	Intended location
Group I	Mines susceptible to firedamp
Group II	Surface industries with explosive gas atmospheres
Group III	Surface industries with explosive dust atmospheres

Group II is subdivided into IIA, IIB and IIC. Group III is subdivided into IIIA combustible flyings, IIIB non-conductive dust and IIIC conductive dust.

TERMINOLOGY TRAP North American Class II means combustible dust. IEC Equipment Group II means surface gas/vapour atmosphere. Class II is not Equipment Group II.

8. Reading complete markings

Division marking

Class I, Division 1, Groups C and D, T4

Part	Meaning
Class I	Flammable gas or vapour hazard
Division 1	Hazard can exist in normal operation, frequently, or under specified combined-failure conditions
Groups C and D	Approved for the stated gas/vapour material groups
T4	Maximum equipment surface temperature associated with T4, normally 135°C

NEC/CSA Zone marking

Class I, Zone 1, AEx db IIC T4 Gb

Part	Meaning
Class I	Gas or vapour hazardous location
Zone 1	Explosive atmosphere likely occasionally in normal operation
AEx	Zone-system explosion-protection marking used in NEC context

Part	Meaning
db	Flameproof protection concept, level b
IIC	Gas subgroup
T4	Temperature class
Gb	High Equipment Protection Level, normally associated with Zone 1

IECEx-style marking

Ex db IIC T4 Gb

Part	Meaning
Ex	Explosion protected
db	Flameproof protection
IIC	Gas subgroup
T4	Temperature class
Gb	Equipment Protection Level

9. How CSA Table 18 applies

Area shown	Visible equipment categories
Zone 0	Zone 0 protection techniques and EPL Ga
Zone 1	Equipment suitable for Zone 0; equipment suitable for Class I, Division 1; Zone 1 techniques; EPL Gb
Zone 2	Equipment suitable for Zone 0; equipment suitable for Zone 1; equipment suitable for Class I, Division 1; equipment suitable for Class I, Division 2; Zone 2 techniques; EPL Gc

10. Practical classification scenarios

Hydrocarbon tank vapour space

Assessment item	Preliminary interpretation
Material	Hydrocarbon vapour
Division material type	Class I
Zone atmosphere type	Gas/vapour, normally IEC Equipment Group II
Occurrence	Continuous, frequent or long duration

Assessment item	Preliminary interpretation
Likely Zone result	Zone 0
Likely Division relationship	Division 1 concept
Normal Zone EPL	Ga
Lesson	Division 1 includes conditions more severe than Zone 1

Routine open sampling point

Assessment item	Preliminary interpretation
Material	Flammable hydrocarbon vapour
Release	Expected occasionally during normal sampling
Likely Zone result	Zone 1 around release
Likely Division result	Division 1
Normal Zone EPL	Gb
Lesson	Division 1 and Zone 1 are broadly similar in this scenario

Outdoor normally tight pump

Assessment item	Preliminary interpretation
Material	Flammable hydrocarbon
Release	Unlikely normally; possible abnormal seal or connection leakage
Ventilation	Good open-air dispersion assumed
Likely Zone result	Zone 2 around credible release sources
Likely Division result	Division 2
Normal Zone EPL	Gc
Lesson	Division 2 and Zone 2 are broadly similar here

Pump in a poorly ventilated acoustic enclosure

Assessment item	Preliminary interpretation
Release	Same pump leakage scenario
Ventilation	Restricted
Accumulation	Vapour can persist inside enclosure
Possible result	More onerous classification or larger extent
Lesson	Do not copy the outdoor classification into the enclosure

Analyser shelter with continuous sample vent

Assessment item	Preliminary interpretation
Inside sample tubing	Hazardous process containment
Around normal analyser vent	Zone 1 or Division 1 type condition

Assessment item	Preliminary interpretation
Shelter interior	Depends on leakage, ventilation, purge and shutdown design
Outside vent termination	Separate hazardous envelope may be required
Lesson	One package can contain multiple classifications

11. Equipment suitability and cross-use

Equipment marking/certification	Zone 0	Zone 1	Zone 2
Ga equipment	Yes	Yes	Yes
Gb equipment	No	Yes	Yes
Gc equipment	No	No	Yes
Class I, Division 1 equipment	Check governing code	May be recognized	May be recognized
Class I, Division 2 equipment	No	Normally no	May be recognized

The Division-certified rows depend on the adopted national installation code and contractual certification requirements. The CSA table supplied by the user visibly recognizes Class I, Division 1 equipment for Zones 1 and 2, and Class I, Division 2 equipment for Zone 2, subject to the applicable Rules and footnotes.

Cross-use decision example

Project requirement: Zone 2, IIB, T3, Gc. Vendor offer: Class I, Division 2, Groups C and D, T3.

1. Confirm that the governing electrical code permits Division 2 equipment in Zone 2.
2. Confirm that the equipment certification covers the actual gas, not only an assumed group comparison.
3. Verify the T3 requirement and certified maximum ambient.
4. Confirm the exact model and certification body approval.
5. Verify cable entries, wiring method and installation instructions.
6. Confirm whether the Purchase Order mandates IECEx or ATEX despite code permissibility.
7. Check client restrictions on mixed certification.
8. Confirm acceptance by the Authority Having Jurisdiction.

CONTRACTUAL POINT An item may be technically acceptable under an installation code but still be contractually noncompliant if the MR or Purchase Order specifically requires IECEx-certified Zone equipment.

12. Project application workflow

Step 1: Establish the governing system

- Project Design Basis and Electrical Design Basis
- Hazardous Area Classification Philosophy
- Client specification
- Material Requisition and datasheet
- Purchase Order
- National electrical code
- Applicable API, IEC, NFPA or CSA reference

Division basis example: API RP 500 / NEC Article 500 / applicable CSA Division rules
 Zone basis example: IEC 60079-10-1 / API RP 505 / NEC Article 505 / applicable CSA Zone rules

Step 2: Record the approved area requirement

Requirement	Example
Classification system	IEC Zone
Zone/Division	Zone 1
Gas/dust group	IIB
Temperature class	T4
EPL	Gb
Ambient range	-10°C to +55°C
Drawing reference	HAC-001 Rev. 3
Applicable code	Project-specified edition

Step 3: Identify the actual material

- Gas composition and worst-case representative gas
- Flash point and vapour pressure
- Auto-ignition temperature
- MESG or material-group classification where applicable
- Dust conductivity and cloud/layer ignition data where applicable

MATERIAL RULE Do not assume every hydrocarbon is Group D or IIA. Use approved process composition and project classification data.

Step 4: Locate every item relative to the envelope

- Motor and space heater
- Local panel and junction boxes
- Solenoids, transmitters and switches
- Vibration and temperature sensors
- Heat tracing and package lighting
- Analyser cabinet, battery or UPS
- Seal-system instruments and vent/drain terminations

Step 5: Read the complete offered marking

Example: Ex db eb IIC T4 Gb

Do not approve a statement such as “suitable for hazardous area” without the complete marking, certificate and schedule.

Step 6: Compare Z-G-T-E-A-C-I

Code	Check	Decision
Z	Zone or Division	Does the equipment suit the classified location?
G	Gas or dust group	Does certification meet the actual material requirement?
T	Temperature class	Is the maximum surface temperature acceptable?
E	EPL	Does the protection level match or exceed the Zone requirement?
A	Ambient range	Does the certificate cover project ambient?
C	Certificate	Is the precise model covered, including X/U conditions?

Code	Check	Decision
I	Installation	Are glands, cable, barriers, earthing and assembly compatible?

Step 7: Review the certificate

- Certification scheme and number
- Exact model and revision
- Complete certificate schedule
- Ambient limitations
- X special conditions and U component status
- Cable-entry conditions and installation instructions

Step 8: Review the complete package

- Mechanical ignition sources and hot process surfaces
- Coupling, guard and static bonding
- Seal leakage, vents and drains
- IS loops and barriers
- Glands and stopping plugs
- Enclosure ventilation or purge
- Package boundary and exact equipment locations

13. Worked package review

Project area requirement: Zone 1, IIB, T4, Gb; maximum ambient 55°C.

Item	Vendor marking / submission
Motor	Class I, Division 1, Groups C and D, T4
Junction box	Ex eb IIC T4 Gb, ambient -40°C to +60°C
Pressure transmitter	Ex ia IIC T4 Ga
Solenoid	Ex ec IIC T4 Gc
Local switch	General-purpose IP66
Coupling	No ignition-hazard assessment submitted

Preliminary multidisciplinary review

Item	Finding
Motor	Potentially usable only if the governing code permits Division 1 equipment in Zone 1, gas coverage is adequate, certification is accepted and the Purchase Order does not mandate IECEx.
Junction box	EPL, gas group, temperature and ambient appear suitable, subject to certificate, terminals and entry-device review.
Pressure transmitter	Ga exceeds the Zone 1 minimum, but the complete intrinsically safe loop must be verified.
Solenoid	Gc is normally inadequate for Zone 1.
Local switch	IP66 does not demonstrate explosion protection; likely unacceptable inside the classified extent.
Coupling	Mechanical ignition risks remain unassessed.
Overall package	Not acceptable as submitted.

Required vendor action

- Provide a suitable Gb solenoid.
- Provide an Ex-certified local switch or relocate it outside the classified extent.
- Submit the motor certificate and jurisdictional/code justification.
- Submit complete certificate schedules and gland details.
- Submit the intrinsically safe loop calculation.
- Submit the mechanical ignition-hazard assessment.
- Revise the package GA to show equipment and classified boundaries.

14. Common mistakes and corrections

Mistake	Why it is wrong	Correct approach
Class I equals Zone 1	Class I identifies gas/vapour material, not occurrence	Add Division or Zone and complete group/T-class data
Division 1 equals Zone 1	Division 1 broadly includes Zone 0 and Zone 1 conditions	Assess frequency and duration under the governing method
Division 2 always equals Zone 2	Broadly similar, but code boundaries and equipment permissions differ	Verify governing code and project basis
Group B always equals IIC	IIC also includes representative Group A gases	Verify actual gas and certificate
Class II equals IEC Group II	Class II is dust; IEC Group II is surface gas	Use the correct terminology
IP66 means Ex	IP addresses ingress, not ignition protection	Require complete Ex certification
IIC alone ensures suitability	EPL, T-class, ambient and installation can still fail	Apply Z-G-T-E-A-C-I
Division and Zone equipment can be freely mixed	Cross-use is jurisdiction- and contract-dependent	Check code permission and PO requirements
Certified motor means certified skid	Other electrical, instrument and mechanical sources remain	Review the complete package

15. One-page memory chart

DIVISION SYSTEM

Class = What type of material?
 Division = How likely is the atmosphere?
 Group = Which gas or dust?
 T-class = Maximum equipment surface temperature

ZONE SYSTEM

Zone = How frequently and for how long?
 Group = Which gas or dust subgroup?
 T-class = Maximum equipment surface temperature
 EPL = Required ignition-protection level

Concept	Simplified relationship
Gas/vapour material	Class I
Occurrence	Division 1 $\approx$ Zone 0 + Zone 1
Lower occurrence	Division 2 $\approx$ Zone 2
Material Group A, acetylene	Broadly IIC
Material Group B, hydrogen	Broadly IIC

Concept	Simplified relationship
Material Group C, ethylene	Broadly IIB
Material Group D, propane	Broadly IIA
Zone EPL	Zone 0 → Ga; Zone 1 → Gb; Zone 2 → Gc

FINAL MEMORY RULE Use Class or Group to identify the hazard type; Division or Zone to identify occurrence; temperature class to control hot surfaces; and EPL or approved Division construction to determine equipment protection.

16. Information Required Before Final Decision

- Country and Authority Having Jurisdiction
- Project hazardous-area classification philosophy
- Governing code and edition
- API RP 500, API RP 505, IEC, NEC or CSA basis
- Purchase Order and MR certification requirements
- Approved HAC drawings and revisions
- Zone or Division for each location
- Gas/dust composition and material group
- Temperature-class requirement
- Required EPL
- Ambient-temperature range
- Complete equipment marking
- Certificate and complete schedule
- X special conditions and U component status
- Cable, gland and wiring method
- Ventilation and purge philosophy
- Package GA and release-source locations
- Approved cross-certification or dual-marking rules
- Client and statutory acceptance requirements

FINAL DECISION OWNERSHIP The responsible discipline engineer, package engineer, process-safety engineer, project authority, client and Authority Having Jurisdiction retain approval responsibility. This guide supports, but does not replace, their decision.

Appendix A. Quick project review form

No.	Review field	Project / vendor information
1	Tag number	
2	Package / equipment	
3	Installation location	
4	HAC drawing and revision	
5	Governing system: Division or Zone	
6	Class, if applicable	
7	Division or Zone	
8	Gas/dust group	
9	Temperature class / maximum surface temperature	

No.	Review field	Project / vendor information
10	Required EPL	
11	Project ambient range	
12	Offered complete marking	
13	Certificate scheme and number	
14	Certificate schedule reviewed	
15	X conditions	
16	U component status	
17	Cable entry / gland / thread	
18	IS loop or purge review	
19	Mechanical ignition-hazard review	
20	PO/MR certification requirement	
21	Preliminary status	
22	Deviation / clarification required	
23	Responsible disciplines	
24	Final approval reference	

Appendix B. Reference framework

Source organization / publication	Application
NFPA 70, National Electrical Code	Hazardous (classified) locations, Division and Zone equipment marking and installation requirements under adopted editions
NFPA 497	Recommended practice for classification of Class I hazardous locations and gas/vapour material properties
API RP 500	Class I, Division 1 and Division 2 petroleum-facility area classification
API RP 505	Class I, Zone 0, Zone 1 and Zone 2 petroleum-facility area classification
IEC 60079-10-1	Classification of areas containing explosive gas atmospheres
IEC 60079-10-2	Classification of areas containing explosive dust atmospheres
IEC 60079-14	Electrical installation design, equipment selection, erection and initial inspection
IECEx resources	Explosion-protection techniques, gas/dust groups and area-classification guidance
CSA C22.1:21, Table 18	User-provided visual reference showing equipment suitable for Zones 0, 1 and 2; licensed publication required for normative application

Always use the project-adopted editions and licensed standards. A newer edition is not automatically contractually applicable. Standard titles are included for identification; this guide does not reproduce normative standard text.